

Q1 extension: How wide should a scoring forecast be?

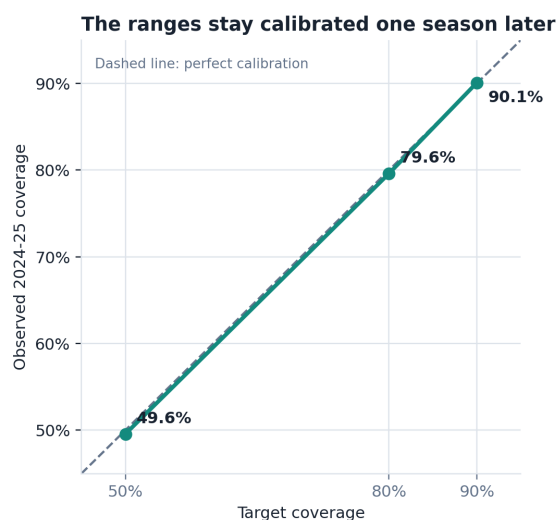
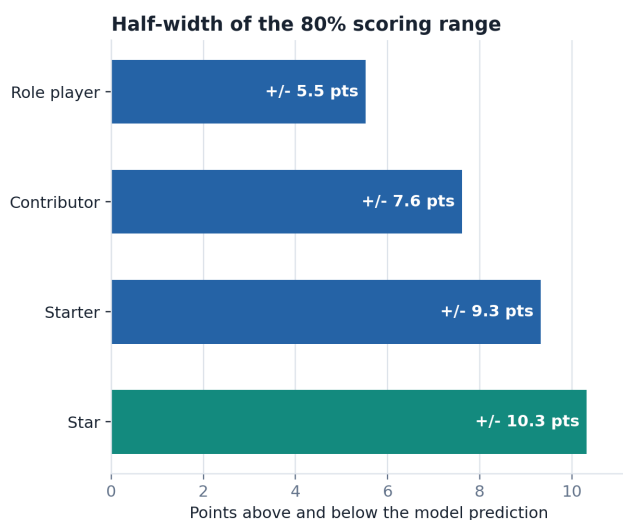
A prediction range shows the uncertainty that a single MAE number hides.

Idea and setup

Published NBA forecast research emphasizes **calibration**: an 80% forecast should contain the real outcome about 80% of the time. We trained Ridge on 2019-20 through 2022-23, used only 2023-24 errors to set the range sizes, and then checked them on 2024-25. This keeps the final season fully unseen when the ranges are chosen.

A useful scoring forecast is a range, not one number

The tested 80% range is almost twice as wide for stars as it is for role players.



Method: Ridge model trained through 2022-23; range sizes learned on 2023-24; final check on 2024-25. Roles use pregame scoring averages.

Figure 1. Left: the number of points added above and below the prediction for an 80% range. Right: target coverage compared with observed coverage on 20,914 future games.

Main result: The global 80% range is +/- 7.25 points and covers 79.6% of the next season. Role players need about +/- 5.5 points, while stars need +/- 10.3. A 25-point star forecast is therefore closer to a range of about 15 to 35 points.

Limitation: these are empirical ranges, not guarantees. They still do not know about late scratches, injuries, lineup announcements, or expected minutes.

Inspiration: Yeh, Rice, and Dubin, Evaluating real-time probabilistic forecasts with application to NBA outcome prediction, arxiv.org/abs/2010.00781. Prediction-interval method inspiration: arxiv.org/abs/1909.07889.

Q2 extension: Why does continuity look so important at first?

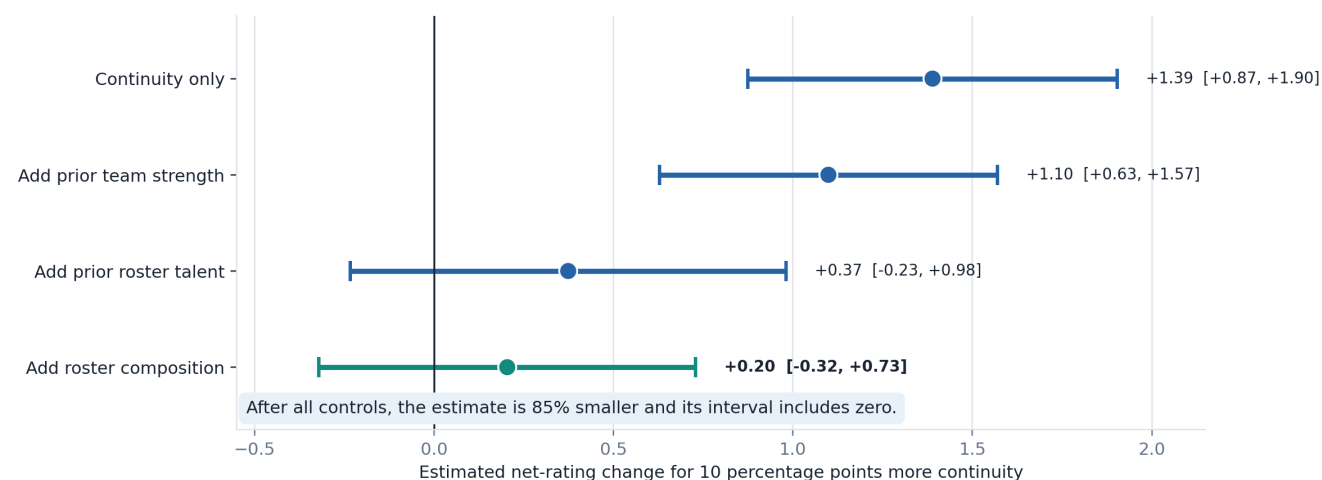
We add the likely confounders one group at a time and watch the estimate change.

Idea and setup

NBA continuity rankings use the percentage of minutes played by returning players. They also note that continuity is related to how good a team already was. We use the same returning-minute idea, then add prior team net rating, prior roster talent, and roster composition. Every model includes season indicators, and the uncertainty is clustered by team.

The continuity effect shrinks after accounting for roster talent

Good teams often keep good players. The raw continuity link includes both effects.



All models include season indicators. Lines are 95% confidence intervals with standard errors clustered by team. Association, not causation.

Figure 2. Each dot is the estimated net-rating difference linked to 10 percentage points more continuity. Lines show 95% confidence intervals. An interval crossing zero means the sample does not separate the estimate from no association.

Main result: The estimate starts at +1.39 net-rating points per 10 percentage points of continuity. After the controls, it falls to +0.20 (95% CI: -0.32 to +0.73, $p = 0.434$). That is an 85% reduction. Much of the raw continuity advantage is really a talent advantage.

Limitation: this is still an observational, full-season measure. It cannot prove that keeping a roster causes better results, and injuries can affect both continuity and performance.

Inspiration: NBA.com, Continuity Rankings, nba.com/news/2025-continuity-rankings. Related definition and team-clustered modeling: Wang, Sarker, and Hosoi (2025), doi.org/10.1177/15270025251328264.

Q3 extension: What actually changes in playoff scoring?

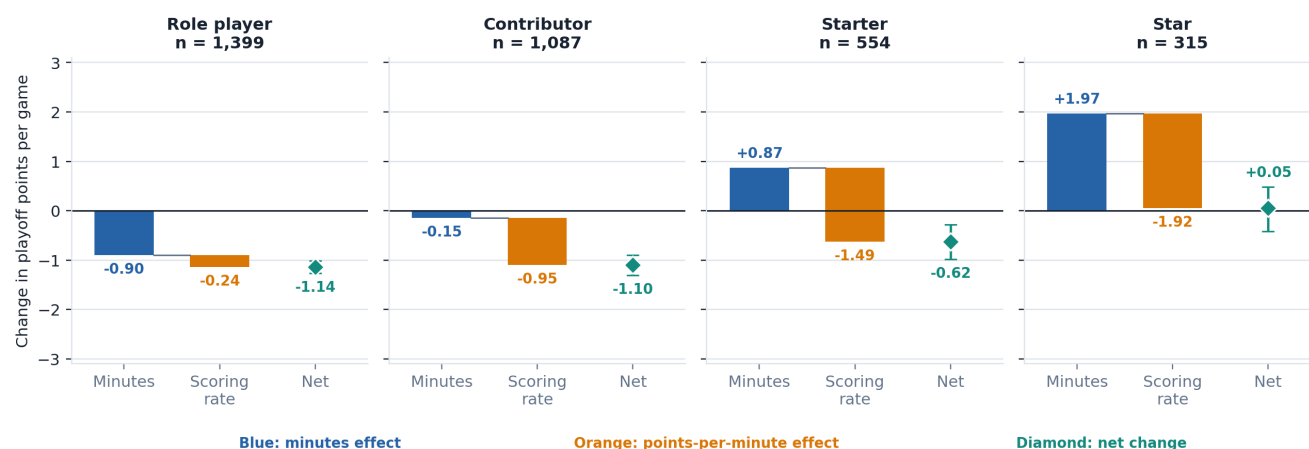
Points per game can change because playing time changes or because scoring rate changes.

Idea and setup

Earlier playoff comparisons often use per-minute efficiency measures so that extra playing time is not mistaken for better performance. We make that idea visual. For every player-season, the change in points per game is split exactly into a minutes effect and a points-per-minute effect. We keep players with at least 20 regular-season games and 5 playoff games.

Stars maintain their scoring because extra minutes offset a lower scoring rate

Each waterfall exactly splits the playoff change in points per game into two parts.



Data: 1995-96 through 2022-23. Minimum 20 regular-season games and 5 playoff games. Net intervals resample players, keeping repeated seasons together.

Figure 3. The first bar shows the scoring change caused by playoff minutes. The second bar adds the change in scoring rate. The diamond is the final change in points per game, with a player-bootstrap 95% confidence interval.

Main result: Across 3,355 player-seasons, scoring falls by 0.93 points per game on average. Most of the drop comes from the scoring-rate effect (-0.84). Stars are different: extra minutes add +1.97 points, while the lower scoring rate removes 1.92. The effects almost cancel, leaving only +0.05 points per game.

Limitation: playoff opponents are stronger and playoff samples are shorter. The chart describes the change but does not identify whether defense, role, health, or coaching caused it.

Inspiration: Game statistics that discriminate winning and losing at the NBA level, pmc.ncbi.nlm.nih.gov/articles/PMC9390892. Related per-minute playoff comparison: FiveThirtyEight, LeBron Doesn't Get Better in the Playoffs. He's Always This Good.